

AI Goes Local

Why the Future of Intelligent Software Runs On-Device

think
tecture

Heads-up

— the benchmark numbers in the following slides reflect the *latest* hardware + software stack updates (post-keynote). They differ from the numbers shown in the keynote recording.

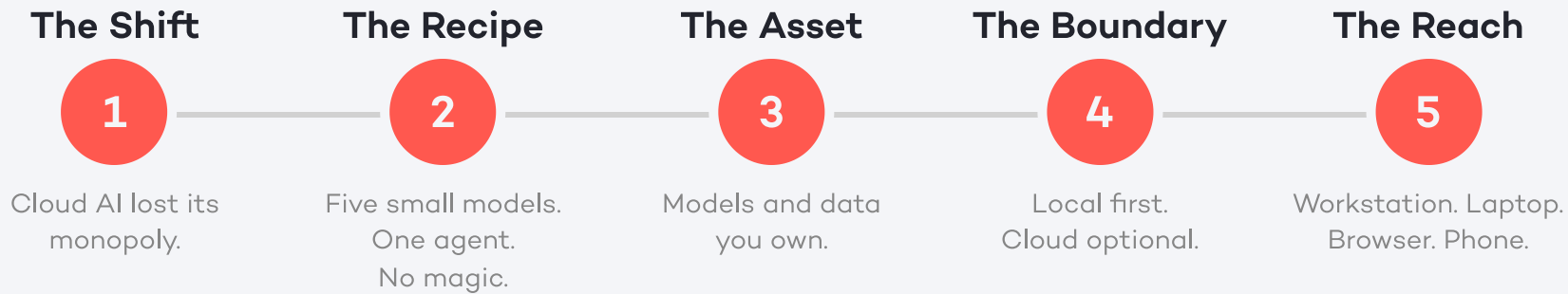
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The Journey



AI-enabled solutions (typically) are

10% AI.

**100% software
engineering.**

PART 1

THE SHIFT

Cloud AI lost its monopoly —
data has gravity, compute is following.

What I Mean by "AI"

WHEN I SAY "AI" TODAY

I DO NOT MEAN

Models

Small. 1–4 B parameters. **Specialised — each does one job.**

Cloud-frontier: GPT-5, Claude, Gemini.

How many

Five, working together. Router · tool-caller · embedder · vision · synthesis.

One giant model, one big API call.

Where it runs

On the laptop on stage. On your phone. In your browser.

On someone else's GPUs.

What it does

Routes, retrieves, generates, sees, listens.

Replaces engineering judgment.

Every demo today is a proof for the left column.

Six Forces Driving Local AI



Privacy — Your data doesn't leave the room.



Regulations — GDPR, HIPAA, DORA, AI Act.



Reliability — No outages. No rate limits. No deprecations.



Latency — Sub-second on a laptop. Cloud round-trip alone is 1–2 s.



Cost — Tokens are metered infrastructure. Local puts you back in control.



Energy — On-device inference is efficient. NPUs sip; data centers gulp.

DEMO #1

Forces — illustrated

Three claims. One query. Watch.



Why Now?



HARDWARE

Consumer GPUs & NPUs and unified memory ship.
\$5k beats what \$50k bought in 2023.

*M5 Max · RTX PRO 6000 · DGX Spark ·
Strix Halo · Qualcomm X Elite
(more coming)*



MODELS

Quality-per-parameter 10×'d.
Sub-4B (partially FT-ed)
reaches what 70B did 18
months ago.

*Gemma 4 · Qwen 3.5/3.6 · LFM 2.5 ·
Mistral Small · GLM (and more)*



ENGINES

Highly optimized code.
Various hardware backends.
Quantization for models & inference.

*llama.cpp · MLX · Transformers.js ·
LEAP SDK (et. al.)*

Three things flipped at once.

And they'll keep flipping. Bet on the architecture!

The Building Blocks



LLM LARGE

Generative text — Cloud frontier models

GPT-5 • Claude • Gemini



SLM SMALL

Generative text — runs on your hardware

Gemma 1-4B • Qwen 4B • LFM 1.2B



VLM

Multimodal — text + image input

Gemma 3 Vision • GLM-V • LFM-VL



Embedding model

Text → vectors for semantic similarity

text-embedding-3-large • EmbeddingGemma



STT

Speech-to-text transcription

Whisper • Parakeet



TTS

Text-to-speech synthesis

Piper • Kokoro



OCR

Document image → text

GLM-OCR • Tesseract

All seven run locally today.

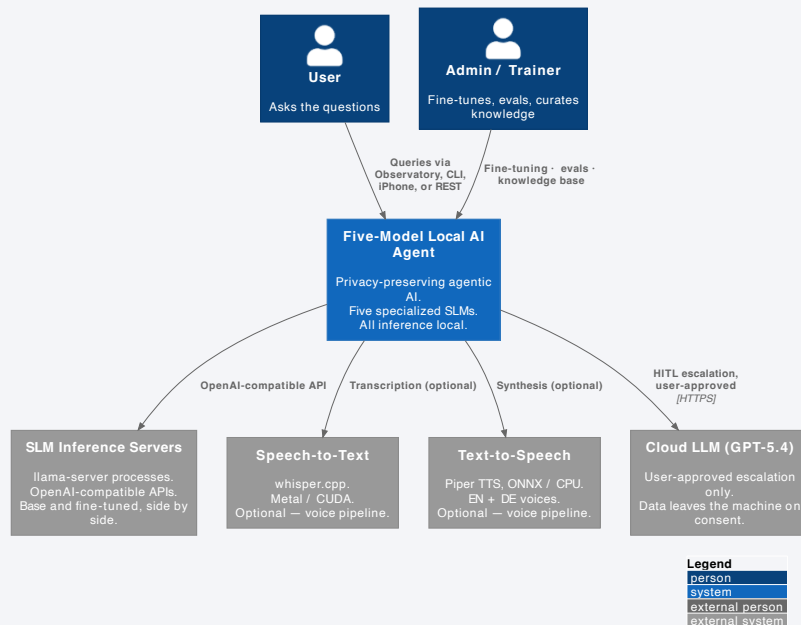
PART 2

THE RECIPE

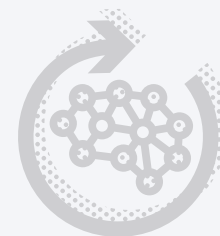
Five small models. One agent. No magic.

This is 'Nextera'

A fictitious SaaS analytics product — running on five small local models.



DEMO #2



The Agent — layer by layer...

Seven capabilities. Five small models. One laptop.

Seven Capabilities

1



Router

LogReg on EmbeddingGemma · intent in ~10 ms · zero LLM calls

2



Tool calling

Qwen 3.5 4B FT · structured JSON: tool + args · WHAT, not HOW

3



Scaffolding

SQLite + Python calculator · deterministic execution · 1 ms

4



Multi-step

Decompose → execute → concretize → calculate → synthesize

5



RAG

EmbeddingGemma vector search · gemma3 4B synthesis

6



Vision

GLM-OCR 0.9B reads pages · pypdf for text · smart hybrid

7



Orchestration

Six layers, four models, one trace — already running on one laptop

65% AI. 35% deterministic non-domain-specific scaffolding.

That's the engineering.

```
# Real components – the deterministic 35%
intent_classifier_logreg.LogRegIntentClassifier # embed + LogReg, ~10 ms total
query_decomposer.MULTI_STEP_PATTERNS          # 16 regex – one step or two?
query_decomposer.decompose                     # JSON parse fallback (line 130)
intent_classifier.looks_like_injection          # 30 regex – prompt-injection guard
sqlite3.execute                                # 1 ms – no model beats stdlib
```

The model decides WHAT. Code does HOW. Both pull their weight.

The 5-Model Stack

- **EmbeddingGemma FT** — Intent via LogReg + Semantic search - **308M**
- **Gemma3 FT** — Query rewrite, tool-result synthesis - **1B**
- **Qwen 3.5 FT** — Tool calling, SQL - **4B**
- **Gemma3** — Vision + RAG document synthesis - **4B**
- **GLM-OCR** — Document reading - **0.9B**

Total: ~10.2 billion parameters.

Less than GPT-3 was in 2020.

The Patterns



RAG

Model finds, not memorizes.

```
embeddings.search(query)
```



Function calling

Model decides WHAT.

Code decides HOW.

```
tools=[sql_query, calc]
```



Multi-step decomposition

Break · Execute · Synthesize.

```
plan → step → step →  
answer
```



Scaffolding

65% AI · 35% deterministic.

```
sql_builder.py · regex
```



Prompt architecture

One prompt per task.

```
prompts/intent.md
```



Scenario-driven config

One JSON file,
zero code changes.

```
scenarios/nextera.json
```

None of these are novel. What's different is — they all run locally.

PART 3

THE ASSET

Models and data you own.

The Fine-Tuning Pipeline



Use

→ Every query logged



Log

→ Structured training data



Train

→ LoRA / QLoRA on your GPU



Deploy

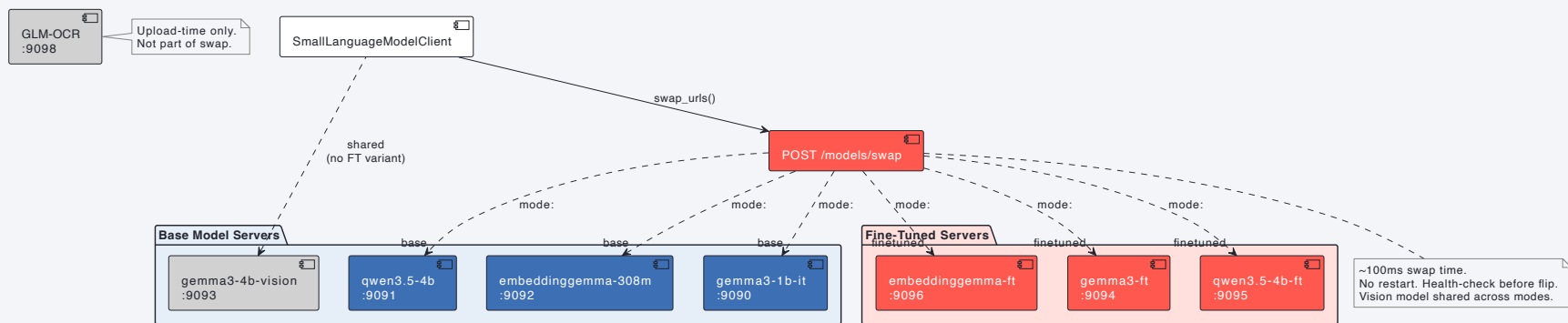
→ Zero-downtime swap

- **SFT** — domain tasks, no RLHF needed
- **LoRA** — ~1% of parameters trained
- **QLoRA** — quantized base, single consumer GPU
- **Tools** — Unsloth · llama.cpp · dual-port
- **Eval gate** — Wilson CI, McNemar's. No vibes.

```
# finetune/train_qwen35_toolcalling.py
peft_config = LoraConfig(
    r=16, lora_alpha=32,
    target_modules=["q_proj", "v_proj"],
    lora_dropout=0.05,
    task_type="CAUSAL_LM",
)
# 1,372 examples · 3 epochs · ~6 min on RTX
```

The model is not a dependency. It's an asset.

Zero-Downtime Swap



```
curl -X POST :8000/models/swap -d '{"mode": "ft"}' → ~100 ms
```

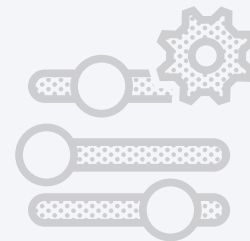
Base on ports 9090–9092. Fine-tuned on 9094–9096.

Atomic flip, no dropped requests.

DEMO #3

The Lobotomy

Without fine-tuning — it just doesn't work.



What Fine-Tuning Bought

Intent classification accuracy

5.6% → 96.7%

Base



5.6%

Fine-tuned



96.7%

Same model. 1,878 fine-tuning examples. ~2 min on RTX.

*Then we replaced the FT model with LogReg —
same data, no LLM call. 99.4%.*

Three Levers: Model · Data · Code

*Domain knowledge lives in three places.
Fine-tuning is just the model lever.*

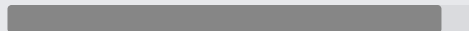


Pick the right model

Qwen 3.5 4B FT v9

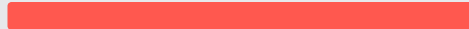
Tool routing (160q)

Old



92.5%

New



99.4%

+6.9 pp

12× fewer errors



Fix the data

47 **corrupted** · 134 **reabeled** · ~10%
of the training set

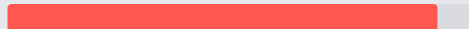
RAG ground-truth, Nextera (80q)

Before



65.0%

After



92.0%

+27 pp

*The third lever - **deterministic code** - you saw live walking the agent.
Knowing which to pull is the engineering.*

Same Architecture. Same Pipeline. Two Domains.

Nextera

SaaS analytics

```
// scenarios/nextera.json (excerpt)
{
  "name": "nextera", "language": "en",
  "paths": { "db": "../data/business.db", ... },
  "models": { "inference_ft": "gemma3-ft-nextera.gguf", ... },
  "sql": { "allowed_tables": ["products", "customers", "sales", "competi"]
}
```

Field Logistics

Public-sector logistics



```
// scenarios/logistics.json (excerpt)
{
  "name": "logistics", "language": "de",
  "paths": { "db": "../data/logistics.db", ... },
  "models": { "inference_ft": "gemma3-ft-logistics.gguf", ... },
  "sql": { "allowed_tables": ["einheiten", "ausrüstung", "ersatzteile", ... ]
}
```

EVAL (RTX, 2026-04-05)

NEXTERA

FIELD LOGISTICS

Intent classification (180q)		93.3%		90.6%
Tool routing (160q)		98.8%		100%
SQL execution		100%		98.8%
RAG ground-truth (80q)		78.8%		70.0%

Same five models. Same pipeline. **One day to retarget.**

Six Industries. One Architecture.

Healthcare — clinical PII, never leaves the device

Finance — structured extraction, audit-grade determinism

Manufacturing — voice on the field tablet, no signal needed

Public sector — sovereignty by construction

Automotive — in-car assistants without the latency tax

Energy — control rooms, no cloud roundtrip

Same pipeline. Same evals. Your domain.

The engineering outlives the models.

PART 4

THE BOUNDARY

Local first. Cloud optional.

When Local. When Cloud.

Pick local when

Data is **regulated, private, or owned by your customer**

Latency matters — sub-second response

Volume is sustained — not occasional, exploratory bursts

Determinism is required — bit-identical reruns

Network can't be guaranteed

Cloud is the right answer when

A **frontier-only** capability is the actual unlock

Privacy and latency aren't your constraints

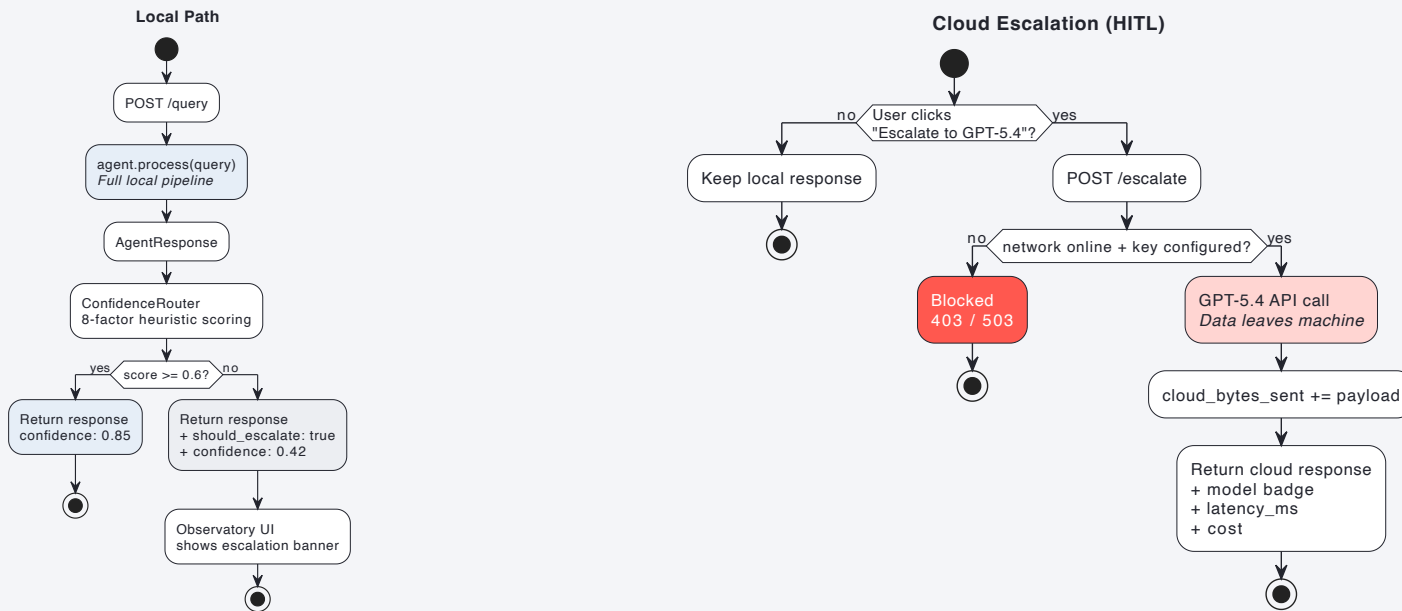
Volume is too small to justify hardware

The task is genuinely **one-off or exploratory**

You need **long-context** — 1M+ tokens

Hybrid is the rule, not the exception.

Hybrid Routing



8-factor score: $< [\text{threshold}] \rightarrow \text{ask the user.}$

Deterministic — no LLM in the loop.

DEMO #4

Local vs. Cloud

Same prompt. Several engines. Read the bytes.



PART 5

THE REACH

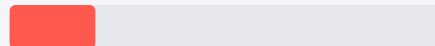
Workstation. Laptop. Browser. Phone.

Four Machines, One Codebase

48 queries · Nextera · bench.py overall median · Q4_K_M synthesis + llama.cpp b9196

NVIDIA RTX PRO 6000

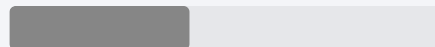
96 GB VRAM · Ubuntu



329 ms

Apple MacBook Pro M5 Max

128 GB unified · macOS



688 ms

AMD Ryzen AI Max+ 395 (Strix Halo)

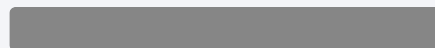
128 GB unified · Fedora



1,455 ms

NVIDIA DGX Spark

128 GB unified · Ubuntu (arm64)

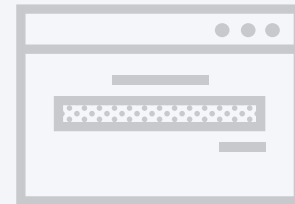


1,657 ms

All four machines re-benched 2026-05-17 — scripts/benchmark.py overall median, 48 queries / FT stack, Q4_K_M synthesis + llama.cpp b9196, byte-deterministic.

Four machines. Four architectures. One codebase.

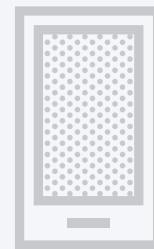
DEMO #5



Browser — zero server, zero network

No backend. No API key. No cloud.

DEMO #6



iPhone — completely offline, in my pocket

Airplane mode. Private data. 1.2B model.

CLOSING

YOUR MOVE

The numbers. The personas.

The Numbers

Revenue analysis · Customer lookups · OCR · Voice · PDFs · Phone offline · Browser zero-server

10.2^B

total params · less than GPT-3 in 2020

4

platforms · Workstation · Laptop · Browser · Phone

~10^{ms}

intent classification · feels instant

~89 tok/s

streaming · in a browser tab

1.2^B

params · running on a phone

0

bytes silently sent · across the session

Take Home



If you're a Developer

Run `llama-server` with a 4B GGUF in 10 minutes. github.com/ggml-org/llama.cpp



If you're an Architect

Map your domain to the three levers — model, data, code. Pick one to start.



If you're a TDM

Pick your most-regulated, highest-volume workload. That's where local pays for itself first.

Three different starting points. One direction.

Models will change. Patterns won't.

DEMO #7

One more thing...

'Your' domain in a couple of seconds.



10% AI. 100% software engineering.

Welcome to the post-cloud era of AI.

Local-first. Cloud-optional. User-controlled.



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